

Network Security and Cryptography

Module 1: Introduction to Network Security

1. Introduction to Security Attacks
 2. Services and Mechanism
 3. Classical encryption techniques
 4. Substitution Ciphers and Transposition ciphers
 5. Cryptanalysis
 6. Steganography
 7. Stream and Block ciphers
 8. A Brief Overview of Hardware Security
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Module 2 : Secret and Public Key Cryptography

1. Modern Block Ciphers: Block Ciphers Principles
2. Shannon's Theory of Confusion and Diffusion
3. Festal Structure
4. Data Encryption Standard (DES)
5. Strength of DES
6. Idea of Differential Cryptanalysis
7. Block Cipher Modes of Operations
8. Triple DES.
9. Advanced Encryption Standard (AES) encryption and decryption
10. Fermat's and Euler's theorem

11. Chinese Remainder theorem
 12. Principals of Public Key Crypto Systems
 13. RSA algorithm
 14. Security of RSA algorithm.
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Module 3 : Authentication Standards & Key Management

1. Message Authentication Codes:
 2. Authentication requirements
 3. authentication functions
 4. message authentication code
 5. Hash Functions
 6. Secure Hash Algorithm (SHA)
 7. Message Digest
 8. Digital Signatures: Digital Signatures
 9. Digital Signature Standards (DSS)
 10. proof of Digital Signature Algorithm
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Module 4: Web Security

1. Authentication Applications: Kerberos
2. Overview of Access Control List (ACL)
3. Electronic Mail Security: Pretty Good Privacy (PGP)

4. S/MIME.
 5. IP Security: Architecture
 6. Authentication Header
 7. Encapsulating Security Payloads
 8. Key Management
 9. Introduction to Secure Socket Layer
 10. Secure electronic, transaction (SET)
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Module 5 : Computational Learning Theory

1. System Security: Introductory Idea of Intrusion
2. Intrusion Detection Techniques
3. Viruses and Worms
4. Firewalls
5. Design Principles of Firewalls
6. IT Acts and Cyber Laws (Global Standards)
7. Virtual Private Network
8. Recent attacks on networks (Packet Analyzer, Buffer Overflow and Port Scanning)
9. Blockchain Technology
10. Preventing Privacy and Piracy on Internet
11. Brief Overview of Cyber Forensic
12. Cyber Forensic Processes
13. Cyber Forensic Applications and Cyber Forensic Tools
14. Digital Evidence

15. Investigating Windows and Unix Systems

16. Email Tracing.